Annotation Instructions: Relevance and Well-Formedness for Multimodal Tip-of-the-Tongue Queries about Scientific Documents

What is ToT

You will be annotating a set of queries about scientific documents. These queries represent questions that a user might ask a search system or post on an online platform when trying to find a ***specific scientific document they have encountered before but can no longer identify***.

The queries describe a partial recollection of the document. The searcher may remember particular aspects of the research, text, figures, tables, graphs, results, methods, or other characteristics, while being uncertain about other details. Because human recollection is imperfect, queries may contain missing information, vague descriptions, uncertainty, or small inconsistencies with the actual document.

Your task is to independently judge:

1. **Relevance:** To what extent does the information in the query align with and identify the presented scientific document?
2. **Well-formedness:** To what extent is the query understandable and usable as a tip-of-the-tongue search query?

Both dimensions are judged independently on a three-point scale: 0, 1, or 2.

Relevance scores:

When assessing relevance, focus exclusively on the information conveyed by the query and its relationship to the document. Do not take stylistic aspects of the query into account when judging relevance. A query may be informal, hesitant, repetitive, grammatically imperfect, or poorly written and still be relevant.

Similarly, a poor or incomplete recollection is not inherently irrelevant. The searcher is explicitly assumed to have forgotten details. A query may therefore contain uncertainty, missing clues, or minor inaccuracies and still receive a high relevance score, provided that the information expressed is generally consistent with and identifies the document.

2: Highly Relevant

The query substantially matches the information described in the paper and represents a plausible recollection of this specific document.
The research, textual, visual, or other clues expressed in the query align with the document.
Minor inaccuracies, uncertainty, omissions, or inconsistencies are acceptable when the overall information clearly points toward the presented paper.
Taking the query as a whole, the information identifies this paper.

1: Partially relevant

The query generally aligns with the information in the paper, but the match is incomplete or uncertain.

Some recollections may be absent from the document, contradicted, inaccurate, or only weakly supported. Nevertheless, when considering all information in the query together, there is meaningful correspondence between the recollection and the document, and the paper is identified to some extent.
Use this category when the document is a plausible match, but the evidence is not sufficiently strong or consistent for a relevance score of 2.

0: Not relevant
The query does not meaningfully align with the presented paper.
The important information or clues expressed in the query are substantially inconsistent with the document, and there is little or no evidence that the searcher could be recalling this paper.
Taking the query as a whole, it is clear that the query does not identify the presented document.
Take care when assigning this annotation to not assign it when the searcher is uncertain or unclear, when there is some identifying information which aligns with the document, assign 1.
As a general guideline, you may assign the following scores based on the percentage of clues that can be tied to the paper:
2 ---> 80%~100% of the elements in the query are aligned with the document
1 ---> 30%~70% of alignment
0 ---> 0~30% of alignment

Well-Formedness

Well-formedness concerns the quality and interpretability of the query as a tip-of-the-tongue request, not whether its recollections are correct.

Do not lower the well-formedness score merely because the query contains factual inaccuracies, imperfect memories, uncertainty, or information that does not match the presented document. These aspects belong to the relevance judgment.

Instead, judge whether a reader or search system can understand what the searcher remembers and what document they are attempting to find.

2 — Well-formed

The query is clear and understandable as a request to identify a scientific document.
The recollected information is expressed coherently enough that the reader can understand the main characteristics of the document being sought. The query may contain uncertainty, informal language, grammatical mistakes, or incomplete recollections, but these do not substantially interfere with understanding.

1: Partially well-formed

The query is understandable overall, but contains noticeable problems that make the intended recollection difficult to interpret.

For example, the query may be unusually vague, fragmented, confusing, internally inconsistent, or poorly structured. Nevertheless, there is still enough understandable information to determine approximately what kind of document the searcher is attempting to find.

0: Not well-formed

The query is so unclear, incomplete, incoherent, or uninterpretable that it is difficult to determine what document the searcher is attempting to describe.
The query provides insufficient understandable information to function meaningfully as a tip-of-the-tongue search request.

Important Annotation Rules

- Annotations must be completed independently and individually.
- Do not discuss individual queries, documents, or annotation decisions with other annotators. Do not compare scores or attempt to reach agreement with another annotator or individuals while performing the task. Uncertainty is a natural part of annotation, please choose the annotation which you feel aligns best.
- Do not strive to assign labels evenly or to assign every label sometimes, there is no relative comparison between queries, and every query should be judged to the requirements listed in this document. It is no problem to assign certain scores often or if you feel that one score fits for no document.
- Do not use ChatGPT, Claude, Gemini, Copilot, or any other language model or AI assistant under any circumstances to interpret a query, inspect a document, verify information, or assist with an annotation decision.
- Do not search for additional information about the paper. Do not use Google, Google Scholar, Semantic Scholar, arXiv, publisher websites, citation databases, or any other external source to identify the document or verify its contents.
- Base your judgment only on the query and the scientific document presented in the annotation interface.
- Judge the query as a whole. Do not require every individual clue to be correct.
- Treat expressions of uncertainty appropriately. Statements such as “I think,” “maybe,” “if I remember correctly,” or “something like” indicate uncertain recollection and should not be interpreted as strict factual claims.
- Missing information is not evidence of irrelevance. A tip-of-the-tongue query is expected to provide only a partial recollection.
- Minor inconsistencies are acceptable for a relevance score of 2 when the overall recollection still clearly identifies the paper.
- Keep relevance and well-formedness separate. A poorly written query may still be highly relevant, and a clearly written query may be completely irrelevant to the presented document.
- Translation is allowed, but judgement must be made on the original sources, for both the query and the document.

Annotation Example
Paper: Debiasing Word Embeddings with Nonlinear Geometry
Query Example 1:

"I'm trying to track down a paper I came across a while back, and I'm hoping someone might recognize it. It involved debiasing of word embeddings and had a fascinating discussion on intersectional biases, particularly focusing on social categories in text data. There was a comparison of multiple bias mitigation methods and an evaluation of bias reduction techniques, with a notable emphasis on the use of nonlinear geometric representations and the identification of bias subspaces. One figure that really stood out was a 3D plot with purple and green shaded regions intersecting, and a black line passing through this intersection, which seemed to illustrate overlapping categories like Gender and Race. I remember there was also a table comparing different models across metrics like Gender, Race, and Religion, with bolded numbers that stood out. I've tried searching with these details but haven't had any luck. Does this ring a bell for anyone? Any help would be greatly appreciated!"

Relevance: 2
The query strongly aligns with the paper. It correctly recalls that the paper concerns debiasing word embeddings, intersectional bias across social categories, nonlinear geometry, and the construction/identification of bias subspaces. The paper specifically studies joint and intersectional biases involving categories such as gender, race, and religion, and proposes JoSEC, which uses nonlinear geometry of individual bias subspaces to construct an intersectional subspace.
The visual recollection is also highly specific and accurate. Figure 2 is a 3D visualization of the Gender and Race bias subspaces, shown as purple and green planes, together with a black line representing their intersection. This is almost exactly what the query describes.
The query also recalls a table comparing approaches across Gender, Race, and Religion. Table 1 contains these three categories and compares several debiasing approaches, with best-performing values bolded. ACL Anthology The paper additionally contains another table evaluating gender, racial, and religious bias in toxicity classification.

Although some descriptions are approximate, for example, referring broadly to a "comparison of multiple bias mitigation methods" rather than naming JoSEC or the specific baselines, these are normal imperfections in a tip-of-the-tongue recollection. The combination of topical, methodological, and highly distinctive visual clues clearly identifies this paper, which is why a relevance score of 2 is assigned.
Well-formedness: 2
The query is clear and understandable as a request to identify a previously encountered scientific paper. It provides several concrete recollections, including the research topic, methodology, social categories, a distinctive figure, and a table. The searcher's uncertainty is expressed naturally and does not make the query difficult to interpret. The query is somewhat verbose, but this is stylistic and does not interfere with its function as a tip-of-the-tongue request.
Example 1 — Relevance: 1, Well-formedness: 1

Query Example 2:
"Looking for a paper about bias word embeddings, where gender and race and maybe religion also. They were trying to remove bias using some geometry thing or spaces, I don't really get how it worked. There was a figure with two things crossing each other, different colors I think, like showing how groups overlap? Also had tables with results of different methods and numbers for the bias. I think they use intersections to do debiasing or maybe

measure it. Not sure how to explain but paper about these bias groups together and embeddings, can anyone help?"
Relevance: 1
The query contains several recollections that correspond to the paper, including debiasing word embeddings, gender and race bias, geometric representations, and intersecting subspaces. However, the descriptions are imprecise and provide limited information about the paper's specific methodology. The document is a plausible match, but the recollections are not sufficiently distinctive or detailed to establish a strong match.
Well-formedness: 1
The query contains noticeable grammatical errors, awkward sentence constructions, and fragmented descriptions. In particular, the relationships between geometric representations, intersecting groups, and the debiasing method are difficult to follow. Nevertheless, the general topic and several recalled characteristics are understandable, making the query usable to some extent.

Query example 3
"paper i saw was about robot or cameras thing? like moving and checking where, also some weather thing maybe. the picture had boxes arrows and like numbers but not the number one, the other ones near graph. was showing how fast? or locations where image go. there was results with something better than other but not sure what was compared. maybe medical image scans not robots. it used model and pictures where the different things are doing different. trying find that one with the shapes and the result part, not sure which."
Relevance: 0
The query describes topics such as robotics, cameras, weather prediction, and medical imaging, which do not correspond to the paper's research on intersectional bias in word embeddings. Its visual and methodological recollections provide no meaningful identifying evidence connecting it to the presented document.
Well-formedness: 0
The query contains substantial grammatical problems, fragmented sentences, unclear references, and disconnected descriptions. Phrases such as "the other ones near graph," "locations where image go," and "different things are doing different" are difficult to interpret. The query does not communicate a coherent research topic or sufficiently understandable set of recollections to determine what scientific document the searcher is seeking.